

AI-103 Practice Exam — Answer Key

Microsoft Certified: Azure AI App and Agent Developer Associate

113 Practice Questions with Correct Answers & Explanations

Sources: open-exam-prep.com + community exam-prep sites (CertyIQ, Pass4Success, ITExams)

Section 1 — Practice Questions (open-exam-prep.com)

Question 1

Which Azure resource is the top-level container in Microsoft Foundry that holds shared infrastructure (storage, key vault, container registry, AI service connections) for one or more AI projects?

✓ A. Microsoft Foundry hub

- B. Azure Cognitive Services account
- C. Azure Resource Group
- D. Azure Subscription

Correct Answer: A. Microsoft Foundry hub

Explanation: A Microsoft Foundry hub is the top-level Azure resource that provides shared infrastructure (storage, key vault, container registry, and connections to AI services) for one or more Foundry projects. Projects are where teams actually build and deploy AI apps and agents.

Question 2

You are building a generative AI app in a Foundry project. Which authentication mechanism is recommended for production code calling Azure OpenAI from Azure-hosted compute?

A. API key embedded in source

✓ B. Microsoft Entra ID with managed identity and the Cognitive Services OpenAI User role

- C. Anonymous access
- D. Shared admin password

Correct Answer: B. Microsoft Entra ID with managed identity and the Cognitive Services OpenAI User role

Explanation: For production, use Microsoft Entra ID with a managed identity assigned to the compute (App Service, Container Apps, Functions, AKS) and grant it the Cognitive Services OpenAI User data-plane role. This implements keyless, RBAC-based access and avoids embedding secrets.

Question 3

Which Azure OpenAI model family is best suited for fast, low-cost summarization of short customer messages where reasoning depth is not required?

A. o3 reasoning models

✓ B. GPT-4.1 mini or gpt-4o-mini

- C. DALL-E 3
- D. Whisper

Correct Answer: B. GPT-4.1 mini or gpt-4o-mini

Explanation: GPT-4.1 mini (or gpt-4o-mini) deliver low latency and low cost per token for routine generation tasks like short summarization where deep multistep reasoning isn't required. The o-series reasoning models are slower and more expensive and better suited for complex multi-step problems.

Question 4

Which Azure OpenAI model class is optimized for complex multistep reasoning, math, code, and chain-of-thought workloads?

- A. Embedding models (text-embedding-3-large)
- ✓ B. o-series reasoning models (e.g., o1, o3)
- C. Whisper
- D. TTS

Correct Answer: B. o-series reasoning models (e.g., o1, o3)

Explanation: Azure OpenAI o-series reasoning models (o1, o3 family) are designed for complex multistep reasoning, math, code, and STEM problems. They use additional inference-time compute and produce higher-quality reasoning at the cost of latency and price.

Question 5

Which Foundry feature provides a low-code visual designer for building, debugging, and evaluating sequences of LLM calls, Python steps, and tool nodes?

- A. Azure Logic Apps
- ✓ B. Prompt flow
- C. Azure Data Factory
- D. Azure Bastion

Correct Answer: B. Prompt flow

Explanation: Foundry prompt flow lets you visually design, debug, and evaluate flows that combine LLM nodes, Python tools, and other connectors. It supports variants, batch evaluation, and tracing, and stores its definition in flow.dag.yaml for source control.

Question 6

What does RAG (retrieval-augmented generation) primarily improve in a generative AI application?

- A. Image generation latency
- ✓ B. Grounding answers in current, organization-specific knowledge that the base model wasn't trained on
- C. GPU utilization on training clusters
- D. Database backup speed

Correct Answer: B. Grounding answers in current, organization-specific knowledge that the base model wasn't trained on

Explanation: RAG retrieves relevant chunks from a knowledge store at query time and supplies them to the LLM as context. This grounds responses in current, organization-specific data that the model never saw during training, reducing hallucinations and enabling accurate domain answers without fine-tuning.

Question 7

Which Azure service is the most common managed vector store used as the retrieval layer for Foundry RAG applications?

- A. Azure SQL Database

✓ **B. Azure AI Search with vector profiles and HNSW indexes**

- C. Azure Files
- D. Azure NetApp Files

Correct Answer: B. Azure AI Search with vector profiles and HNSW indexes

Explanation: Azure AI Search is the first-class managed vector store for Foundry RAG. It supports HNSW vector indexes, hybrid (keyword + vector) search, semantic ranking, security trimming, and integrated vectorization, making it well-suited for production RAG.

Question 8

You want Azure AI Search to automatically chunk documents and generate embeddings during indexing. Which feature should you enable?

- A. Manual ingestion only

✓ **B. Integrated vectorization (skillsets with embedding skill and split skill)**

- C. Geo-replication
- D. Full-text only

Correct Answer: B. Integrated vectorization (skillsets with embedding skill and split skill)

Explanation: Azure AI Search integrated vectorization configures a skillset that splits documents into chunks and calls an embedding model (e.g., Azure OpenAI text-embedding-3) to populate vector fields automatically during indexing. This eliminates the need to write custom ingestion code.

Question 9

Which retrieval strategy in Azure AI Search combines BM25 keyword search and vector similarity, then re-ranks the merged results using a Microsoft cross-encoder model?

- A. Pure keyword

✓ **B. Hybrid search with semantic ranker**

- C. Geo-spatial search
- D. Faceted search

Correct Answer: B. Hybrid search with semantic ranker

Explanation: Hybrid search runs BM25 and vector queries in parallel and fuses results. When semantic ranking is enabled, the top results are re-ordered by a Microsoft cross-encoder model. This combination typically outperforms keyword-only or vector-only retrieval for RAG.

Question 10

Which Azure OpenAI capability lets a model decide to invoke a developer-defined function (tool) and have the host app execute it before returning a final answer?

✓ **A. Function calling (tool calling)**

- B. Embeddings
- C. Whisper
- D. Content filters

Correct Answer: A. Function calling (tool calling)

Explanation: Function calling (also called tool calling) lets you describe one or more functions to the model. The model can then return a structured request to invoke a function with arguments. The host app executes the function and returns the result so the model can produce the final answer.

Question 11

Which Microsoft Foundry capability provides server-side state, threads, runs, and tool orchestration so you don't have to manage chat history yourself when building an agent?

- A. Azure DevOps boards
- ✓ B. Microsoft Foundry Agent Service / Assistants API
- C. Azure Service Bus queues
- D. Azure Notification Hubs

Correct Answer: B. Microsoft Foundry Agent Service / Assistants API

Explanation: Foundry Agent Service (and the underlying Assistants-style API) manages threads, runs, message history, and tool execution server-side. You define an agent with instructions, tools, and a model; the service handles persistent conversation state and tool orchestration.

Question 12

Which built-in tool can be attached to a Foundry agent so it can answer questions grounded in a specific Azure AI Search index without you writing retrieval code?

- ✓ A. Azure AI Search tool (file search / index grounding)
- B. Azure DNS tool
- C. Azure Backup tool
- D. Azure Front Door tool

Correct Answer: A. Azure AI Search tool (file search / index grounding)

Explanation: Foundry agents support a built-in Azure AI Search tool (also referred to as file search or grounding) that lets the agent retrieve relevant content from a configured index automatically. The agent invokes the tool when it determines retrieval would help answer the user's question.

Question 13

Which framework is Microsoft's recommended option for building multi-agent orchestrations in .NET and Python that integrate with Foundry?

- ✓ A. Microsoft Agent Framework (and/or Semantic Kernel)
- B. Apache Spark
- C. TensorFlow
- D. Azure Service Fabric

Correct Answer: A. Microsoft Agent Framework (and/or Semantic Kernel)

Explanation: Microsoft Agent Framework (which builds on Semantic Kernel concepts) is the recommended .NET and Python framework for building agents and multi-agent orchestrations on Azure. It integrates with Foundry models, prompt flow, and the Foundry Agent Service.

Question 14

Which Azure AI Vision capability extracts printed and handwritten text from images and PDFs?

- A. Image captioning
- ✓ B. Read API (OCR)
- C. Face detection
- D. Speaker recognition

Correct Answer: B. Read API (OCR)

Explanation: The Read API in Azure AI Vision performs OCR on printed and handwritten text in images and PDFs. It returns text, bounding boxes, language, and confidence scores. For richer document understanding (forms, tables, structured fields), use Azure AI Document Intelligence.

Question 15

You need to extract structured fields (vendor name, total, invoice number, line items) from invoices in many formats. Which Azure service provides a prebuilt model for this?

✓ A. Azure AI Document Intelligence prebuilt invoice model

- B. Azure AI Search default analyzer
- C. Azure Cognitive Services Speech
- D. Azure Cosmos DB analytical store

Correct Answer: A. Azure AI Document Intelligence prebuilt invoice model

Explanation: Azure AI Document Intelligence ships a prebuilt invoice model that extracts vendor, total, invoice number, line items, and many other common invoice fields out of the box. For specialized formats you can train custom extraction or classification models.

Question 16

Which Azure AI Speech capability converts continuous audio into text, supports multiple languages, and offers custom acoustic and language models?

✓ A. Speech to text

- B. Text to speech
- C. Speech translation
- D. Voice live

Correct Answer: A. Speech to text

Explanation: Azure AI Speech's speech-to-text service transcribes audio into text in many languages and supports custom speech models tuned with your domain audio and lexicon. Text to speech and speech translation are different services within the Speech API family.

Question 17

Which Azure AI Speech feature lets you control pronunciation, pauses, prosody, and voice style in synthesized output using XML-based markup?

A. JSON config only

✓ B. Speech Synthesis Markup Language (SSML)

- C. ARM templates
- D. OpenAPI

Correct Answer: B. Speech Synthesis Markup Language (SSML)

Explanation: SSML (Speech Synthesis Markup Language) is the W3C XML-based markup that Azure AI Speech text-to-speech accepts to control pronunciation, pauses, prosody, voice, role, style, and degree. Using SSML enables natural and brand-aligned voice output.

Question 18

Which Azure AI Language feature lets you build a conversational understanding model that classifies user utterances into intents and extracts entities?

✓ A. Conversational Language Understanding (CLU)

- B. Custom Vision
- C. Speaker Recognition
- D. Anomaly Detector

Correct Answer: A. Conversational Language Understanding (CLU)

Explanation: Conversational Language Understanding (CLU) in Azure AI Language is the modern replacement for LUIS. It lets you train a model on labeled utterances to classify intents and extract entities, supporting multi-turn and multi-language scenarios.

Question 19

Which Azure AI Language capability lets you build a question-answering knowledge base from FAQ pages, manuals, or URLs with multi-turn follow-up support?

✓ **A. Question Answering (custom QnA)**

- B. Translator
- C. Speech to text
- D. Personalizer

Correct Answer: A. Question Answering (custom QnA)

Explanation: The Question Answering feature in Azure AI Language ingests FAQ pages, URLs, and documents to build a knowledge base, supports multi-turn follow-up prompts, and can be deployed as a REST endpoint. It is the modern replacement for QnA Maker.

Question 20

Which Azure AI Content Safety category specifically detects user prompts attempting to bypass model safety guardrails (jailbreak attempts)?

A. Hate severity

✓ **B. Prompt Shields (direct and indirect attack detection)**

- C. Sexual severity
- D. Violence severity

Correct Answer: B. Prompt Shields (direct and indirect attack detection)

Explanation: Azure AI Content Safety includes Prompt Shields, which detect direct jailbreak attempts and indirect (cross-prompt) injection attempts in retrieved content. The harm category models (hate, sexual, violence, self-harm) detect harmful content rather than attack patterns.

Question 21

Which built-in Foundry evaluator measures whether a generated response is supported by the retrieved context in a RAG application?

A. Coherence

B. Fluency

✓ **C. Groundedness**

D. Latency

Correct Answer: C. Groundedness

Explanation: Groundedness measures the degree to which the generated response is supported by the supplied context. It is a core RAG-quality evaluator. Coherence and fluency assess language quality without reference to context, and latency is a performance metric.

Question 22

Which Foundry feature lets you perform A/B testing of two prompt variants on a fixed evaluation dataset and compare quality metrics?

A. Manual chat in playground

✓ **B. Bulk test (batch run) with prompt variants and built-in evaluators**

C. Azure DDoS Protection

D. Azure Cost Management

Correct Answer: B. Bulk test (batch run) with prompt variants and built-in evaluators

Explanation: A bulk test (batch run) executes a flow against many rows of an evaluation dataset for each variant, with built-in evaluators (groundedness, relevance, coherence, fluency) yielding per-variant metrics. This supports objective A/B comparison of prompts.

Question 23

Which Python SDK is the modern unified entry point for working with Foundry projects, including agents, deployments, evaluations, and connections?

✓ **A. azure-ai-projects**

B. azure-storage-blob

C. azure-cosmos

D. azure-mgmt-network

Correct Answer: A. azure-ai-projects

Explanation: azure-ai-projects is the unified Python SDK for Foundry projects. It exposes clients for agents, deployments, evaluations, datasets, indexes, connections, and more. azure-ai-inference is its companion for direct model inference.

Question 24

Which Azure SDK package provides a unified inference client that calls deployed Foundry models (Azure OpenAI, open-source, multimodal) via a single API surface?

✓ **A. azure-ai-inference**

B. azure-iot-hub

C. azure-monitor-query

D. azure-mgmt-resource

Correct Answer: A. azure-ai-inference

Explanation: azure-ai-inference provides a unified Python (and .NET) inference client that targets deployed Foundry models with a single API, regardless of whether the model is Azure OpenAI, an open-source serverless API, or a managed-compute deployment. It supports chat, embeddings, and image inputs.

Question 25

Which content filter setting in an Azure OpenAI deployment determines the severity threshold for hate, sexual, violence, and self-harm categories?

A. TLS protocol version

✓ **B. Per-category severity (low, medium, high) for both prompt and completion**

C. Region of the deployment

D. Model name

Correct Answer: B. Per-category severity (low, medium, high) for both prompt and completion

Explanation: Azure OpenAI deployments allow you to configure a content filter per deployment with severity thresholds (low, medium, high) for hate, sexual, violence, and self-harm categories applied to both prompts and completions, plus optional Prompt Shields and protected material detection.

Question 26

Which Azure AI Vision feature analyzes video streams for events such as people counting and zone violations in retail or industrial scenarios?

A. Speaker Recognition

✓ **B. Spatial Analysis**

C. Translator

D. QnA Maker

Correct Answer: B. Spatial Analysis

Explanation: Spatial Analysis in Azure AI Vision processes video to count people, detect zone entry and exit, social distancing, face mask detection, and other events. It is typically deployed as containers near the camera for low latency.

Question 27

Which Azure service indexes video content to enable searchable transcripts, face recognition, OCR, sentiment, and topic extraction?

✓ **A. Azure AI Video Indexer**

B. Azure Front Door

C. Azure Container Registry

D. Azure Service Bus

Correct Answer: A. Azure AI Video Indexer

Explanation: Azure AI Video Indexer extracts insights from video and audio: transcript, translation, OCR, face recognition, speaker identification, topics, keywords, sentiment, and brand detection, exposing them via REST APIs and a UI for search and discovery.

Question 28

Which Foundry capability deploys a foundation model on dedicated managed VMs inside your VNet, suitable for open-source or custom models that need network isolation?

A. Standard pay-as-you-go API

✓ **B. Managed compute deployment**

C. Anonymous endpoint

D. Free tier

Correct Answer: B. Managed compute deployment

Explanation: Foundry managed compute deployments host a model on Azure-managed dedicated GPU VMs, with optional VNet integration and private endpoints. This is appropriate for open-source models, custom fine-tuned models, or workloads that require network isolation and predictable performance.

Question 29

Which approach reserves dedicated capacity in Azure OpenAI for a high-volume workload to guarantee throughput and latency?

A. Increase the chat temperature

✓ **B. Provisioned Throughput Units (PTUs)**

C. Disable rate limits

D. Use embeddings only

Correct Answer: B. Provisioned Throughput Units (PTUs)

Explanation: Provisioned Throughput Units (PTUs) reserve dedicated model capacity for a deployment, providing guaranteed throughput and stable latency. This contrasts with the standard pay-as-you-go option, which shares capacity and can be rate-limited under load.

Question 30

Which prompt-engineering technique gives the LLM a few input/output examples directly in the prompt so it generalizes the pattern?

- A. Zero-shot prompting
- ✓ B. Few-shot prompting
- C. Fine-tuning
- D. Chain-of-thought distillation

Correct Answer: B. Few-shot prompting

Explanation: Few-shot prompting includes a small number of input/output examples in the prompt, teaching the model the desired pattern in-context. Zero-shot omits examples; fine-tuning updates the model itself; chain-of-thought elicits step-by-step reasoning.

Question 31

Which Azure OpenAI capability allows you to fine-tune a chat model (e.g., GPT-4.1 mini) with curated input/output pairs to improve performance on a specific task?

- A. Embeddings
- ✓ B. Fine-tuning (supervised) on the Azure OpenAI service
- C. Whisper
- D. Image generation

Correct Answer: B. Fine-tuning (supervised) on the Azure OpenAI service

Explanation: Azure OpenAI supports supervised fine-tuning of selected chat and embedding models with curated input/output pairs in JSONL format. After training, the fine-tuned model is deployed like any other Azure OpenAI deployment and can be called via the standard SDKs.

Question 32

Which Azure OpenAI embedding model is recommended in 2026 for general-purpose semantic search and retrieval?

- ✓ A. text-embedding-3-large (or text-embedding-3-small for cheaper use cases)
- B. DALL-E 3
- C. Whisper
- D. TTS

Correct Answer: A. text-embedding-3-large (or text-embedding-3-small for cheaper use cases)

Explanation: text-embedding-3-large is Azure OpenAI's high-quality general-purpose embedding model. text-embedding-3-small is a lower-cost option for less demanding use cases. Both produce vectors usable for semantic search, clustering, and recommendations.

Question 33

What is a key responsible-AI risk you must mitigate when deploying a generative AI assistant that retrieves from user-uploaded content?

- ✓ A. Prompt injection from malicious documents that try to redirect the assistant
- B. GPU thermal throttling
- C. TCP packet loss
- D. Region failover lag

Correct Answer: A. Prompt injection from malicious documents that try to redirect the assistant

Explanation: Indirect prompt injection occurs when malicious instructions are embedded in retrieved content (web pages, uploaded files, emails). Mitigations include Azure AI Content Safety prompt shields, hardened system prompts that treat retrieved content as untrusted, and tool-call guardrails.

Question 34

Which Foundry capability enables continuous monitoring of a deployed generative AI app for groundedness, relevance, and safety on a sample of production traffic?

✓ A. Continuous evaluation in Foundry monitoring

- B. Azure Backup
- C. Azure Site Recovery
- D. Azure Bastion

Correct Answer: A. Continuous evaluation in Foundry monitoring

Explanation: Foundry continuous evaluation samples production requests/responses and applies configured evaluators (groundedness, relevance, content safety, custom) on a recurring schedule. This shifts evaluation from a pre-release gate to ongoing observability.

Question 35

Which Azure service provides per-deployment metrics (request count, latency, prompt tokens, completion tokens, content-filter triggers) for an Azure OpenAI deployment?

✓ A. Azure Monitor (Azure OpenAI metric namespace)

- B. Azure DDoS Protection
- C. Azure Migrate
- D. Azure Backup

Correct Answer: A. Azure Monitor (Azure OpenAI metric namespace)

Explanation: Azure OpenAI emits metrics to Azure Monitor in its dedicated namespace, including per-deployment requests, latency, prompt and completion tokens, and content-filter triggers. Application Insights and Log Analytics can be used to build dashboards and alerts.

Question 36

Which agent design pattern uses one orchestrator agent that delegates subtasks to specialized worker agents and aggregates their results?

A. Single-agent chat

✓ B. Hierarchical multi-agent (orchestrator + workers)

- C. Round-robin polling
- D. Pure broadcast

Correct Answer: B. Hierarchical multi-agent (orchestrator + workers)

Explanation: A hierarchical multi-agent design has a top-level orchestrator agent that decomposes tasks and delegates them to specialized worker agents (e.g., research, coding, summarization), then aggregates the results. Microsoft Agent Framework supports this and other multi-agent topologies.

Question 37

Which Foundry capability ensures that an agent's tool calls are restricted to a defined allowed list (so injected instructions cannot trigger arbitrary tools)?

✓ A. Tool whitelisting in the agent definition (the agent only exposes the tools you register)

- B. Disable Azure Monitor
- C. Use plaintext keys
- D. Enable anonymous access

Correct Answer: A. Tool whitelisting in the agent definition (the agent only exposes the tools you register)

Explanation: Agents can only call tools you explicitly register with the agent. Restricting the agent to the minimum required toolset (and applying authorization in the tool implementations themselves) limits the blast radius of any prompt-injection or model misbehavior.

Question 38

Which Azure AI Language feature translates text between many languages and is recommended over standalone Translator for new development inside Foundry workflows?

✓ **A. Translator (Azure AI Translator)**

- B. Speech to text
- C. Custom Vision
- D. Personalizer

Correct Answer: A. Translator (Azure AI Translator)

Explanation: Azure AI Translator (part of the Azure AI services / Language family) supports text translation across many languages, document translation, and custom translation (Custom Translator). It is the recommended Azure-native translation API for app developers.

Question 39

Which Foundry concept stores a named, reusable connection (with credentials) from a project to an external service like Azure OpenAI or Azure AI Search?

✓ **A. Connection**

- B. Datastore
- C. Endpoint
- D. Compute target

Correct Answer: A. Connection

Explanation: A Foundry connection is a named, reusable reference to an external service plus its authentication method (managed identity, API key, Entra ID). Flows, code, and agents consume connections by name without embedding credentials.

Question 40

Which approach is recommended to reduce token cost in a RAG application without significantly degrading quality?

A. Always pass the entire knowledge base in the system prompt

✓ **B. Improve retrieval precision so fewer, more relevant chunks are sent to the model**

- C. Set max_tokens to a very large value
- D. Disable embeddings

Correct Answer: B. Improve retrieval precision so fewer, more relevant chunks are sent to the model

Explanation: Token cost is dominated by prompt length, which is dominated by retrieved context. Improving retrieval precision (better embeddings, hybrid search, semantic re-ranking, smaller k, tighter chunking) reduces context size while preserving or improving quality.

Question 41

Which Azure AI Search feature filters retrieval results based on a user's group membership so users see only documents they are authorized to view?

✓ **A. Security trimming with filter on principal IDs**

- B. TLS termination
- C. Geo-redundant storage
- D. Content moderation

Correct Answer: A. Security trimming with filter on principal IDs

Explanation: Azure AI Search supports security trimming by storing principal IDs (e.g., Entra ID groups) on each document and filtering at query time using filter expressions. This ensures users only see documents whose principal list intersects with their group memberships.

Question 42

Which Azure AI Document Intelligence model type would you use to extract custom fields (e.g., 'student ID', 'enrollment date') from a school enrollment form?

✓ A. Custom extraction model trained on labeled samples

- B. Prebuilt invoice model
- C. Read API
- D. Translator

Correct Answer: A. Custom extraction model trained on labeled samples

Explanation: When fields are domain-specific and not covered by a prebuilt model, train a custom extraction model in Azure AI Document Intelligence with labeled samples. The trained model can be deployed and called from your application to extract those fields from new documents.

Question 43

Which multimodal capability of Azure OpenAI gpt-4o lets a single model accept image and text inputs in the same request?

✓ A. Vision-enabled chat completions

- B. Function calling only
- C. Embeddings only
- D. Speech-to-text only

Correct Answer: A. Vision-enabled chat completions

Explanation: gpt-4o (and successors) accept multimodal chat completion inputs that combine text and image content parts in the same request. This enables tasks like answering questions about a screenshot, describing a chart, or summarizing a document image, in one model call.

Question 44

Which Azure OpenAI feature streams generated tokens to the client as they're produced, reducing time-to-first-token in chat UIs?

✓ A. Server-sent events streaming (stream=true) in chat completions

- B. Batch transcription
- C. Cold storage
- D. Synchronous polling once per minute

Correct Answer: A. Server-sent events streaming (stream=true) in chat completions

Explanation: Setting stream=true on Azure OpenAI chat completions returns a stream of server-sent events containing partial tokens as they're generated. This dramatically improves perceived latency in chat UIs by letting users start reading immediately.

Question 45

Which feature of structured outputs in Azure OpenAI ensures the model returns JSON that conforms to a developer-supplied schema?

- A. Free-form prose only
- ✓ **B. JSON schema-constrained structured outputs**
- C. TLS
- D. Embeddings

Correct Answer: B. JSON schema-constrained structured outputs

Explanation: Structured outputs constrain the model to emit JSON conforming to a developer-supplied JSON schema. This eliminates parsing errors and is ideal for tool calls, function arguments, and downstream automation that expects strict data shapes.

Question 46

Which Foundry feature enables you to ground a chat agent in a connected Azure AI Search index by attaching it as a data source to an Azure OpenAI deployment?

- ✓ **A. On Your Data (Azure OpenAI On Your Data)**
- B. Azure DNS records
- C. Azure Bastion
- D. Azure Front Door

Correct Answer: A. On Your Data (Azure OpenAI On Your Data)

Explanation: Azure OpenAI On Your Data (the data sources feature) lets you attach an Azure AI Search index (or other supported sources) directly to a chat completion call. The service handles retrieval, prompt construction, and citation generation.

Question 47

Which Azure AI Content Safety classification protects against the model generating verbatim copyrighted content (e.g., song lyrics, code)?

- A. Hate severity
- ✓ **B. Protected material detection**
- C. Sexual severity
- D. Self-harm severity

Correct Answer: B. Protected material detection

Explanation: Azure AI Content Safety includes protected material detection that identifies generated content matching known copyrighted material such as song lyrics, code, and selected text. It can be enabled per Azure OpenAI deployment as part of the content filter.

Question 48

Which Foundry construct represents a deployed model endpoint that an app or agent can call, distinct from the underlying model artifact?

- ✓ **A. Deployment**
- B. Hub
- C. Registry
- D. Subscription

Correct Answer: A. Deployment

Explanation: A deployment in Foundry / Azure OpenAI is the named, callable instance of a specific model and version with its own throughput, content filter, and (optionally) PTU/managed-compute configuration. Multiple deployments can wrap the same underlying model.

Question 49

Which technique should you use to limit the maximum length of an LLM response in chat completions?

✓ A. max_tokens parameter (or max_completion_tokens)

- B. Increase temperature
- C. Set top_p to 1
- D. Use a longer system prompt

Correct Answer: A. max_tokens parameter (or max_completion_tokens)

Explanation: max_tokens (or the newer max_completion_tokens for reasoning models) caps the number of tokens the model can generate in the response. This protects against runaway responses, controls cost, and helps fit responses into downstream constraints.

Question 50

Which sampling parameter controls how deterministic vs. creative an LLM's output is, with 0 being most deterministic?

✓ A. temperature

- B. frequency_penalty only
- C. stop sequences
- D. max_tokens

Correct Answer: A. temperature

Explanation: temperature controls randomness of the output; near 0 is deterministic and useful for extraction or classification, higher values (e.g., 0.7-1.0) are more creative and useful for ideation. top_p is an alternative nucleus sampling control.

Question 51

Which Foundry capability provides distributed tracing of every step in a prompt flow (retrieval, prompt build, LLM call, tool invocation) using OpenTelemetry?

✓ A. Tracing in Foundry observability (OpenTelemetry-compatible)

- B. Azure Service Health
- C. Azure Cost Alerts
- D. Azure Backup

Correct Answer: A. Tracing in Foundry observability (OpenTelemetry-compatible)

Explanation: Foundry instruments prompt flows and SDK calls with OpenTelemetry-compatible distributed tracing. Each request produces a trace with spans for retrieval, prompt construction, LLM calls, and tool invocations, surfaced in Application Insights and Foundry's tracing UI.

Question 52

Which Azure feature lets you alert when an Azure OpenAI deployment's daily token usage exceeds a threshold?

✓ A. Azure Monitor metric alert on token-usage metrics with Action Group notification

- B. Azure Bastion
- C. Azure Site Recovery
- D. Azure Migrate

Correct Answer: A. Azure Monitor metric alert on token-usage metrics with Action Group notification

Explanation: Azure Monitor metric alerts can fire on Azure OpenAI's prompt and completion token metrics with thresholds and action groups (email, webhook, Logic Apps). This is the recommended way to detect and react to runaway token usage.

Question 53

Which built-in agent tool lets a Foundry agent execute Python code in a sandboxed interpreter (for analysis, math, or chart generation)?

✓ **A. Code Interpreter**

- B. Logic Apps connector
- C. Azure DNS query tool
- D. Azure Front Door tool

Correct Answer: A. Code Interpreter

Explanation: The Code Interpreter tool runs Python in a sandboxed environment as part of an agent's tool call. The agent can analyze uploaded files, perform calculations, generate plots, and return results back into the conversation.

Question 54

Which agent feature lets you upload reference documents (PDFs, DOCX) and have the agent answer grounded in them without you provisioning an Azure AI Search index?

✓ **A. File Search (built-in agent file grounding)**

- B. Azure Storage soft delete
- C. Azure Cosmos DB throughput
- D. Azure CDN cache

Correct Answer: A. File Search (built-in agent file grounding)

Explanation: The File Search tool lets you attach files to an agent or thread; the service handles chunking, embedding, and retrieval automatically, so the agent can answer questions grounded in those files without you wiring up Azure AI Search yourself.

Question 55

Which design choice helps ensure that retrieved citations are surfaced to the user so they can verify the source of each grounded statement?

A. Hide all sources

✓ **B. Use Azure OpenAI On Your Data or a flow that returns citation metadata, and render citations in the UI**

- C. Use temperature 1.5
- D. Set max_tokens to 8

Correct Answer: B. Use Azure OpenAI On Your Data or a flow that returns citation metadata, and render citations in the UI

Explanation: Azure OpenAI On Your Data and well-designed RAG flows return citation metadata alongside the answer, allowing the UI to render footnotes/links to source chunks. This builds user trust and supports verification, a responsible-AI best practice.

Question 56

Which Azure OpenAI feature lets you batch many independent prompts into a single asynchronous job that completes within 24 hours at a discounted price?

A. Real-time streaming chat

✓ **B. Batch API (asynchronous batch processing)**

- C. On Your Data
- D. Embeddings only

Correct Answer: B. Batch API (asynchronous batch processing)

Explanation: The Azure OpenAI Batch API accepts a JSONL file of independent requests and processes them asynchronously, typically within 24 hours, at significantly reduced per-token cost. It's well-suited to bulk classification, summarization, or evaluation jobs.

Question 57

Which Azure compute option is most appropriate for hosting a stateless web app that serves a Foundry-powered chat UI with autoscale and easy continuous deployment?

✓ A. Azure App Service or Azure Container Apps

- B. Azure HPC dedicated bare metal
- C. Azure Batch
- D. Azure Sphere

Correct Answer: A. Azure App Service or Azure Container Apps

Explanation: Azure App Service and Azure Container Apps are managed compute options that support autoscaling, managed identities, and continuous deployment, making them ideal for stateless chat UIs that call Foundry. AKS is also valid but adds operational complexity.

Question 58

Which strategy reduces hallucinations in a RAG-grounded chatbot the most?

- A. Increase max_tokens

✓ B. Improve retrieval quality (chunking, embeddings, hybrid search) and tighten the system prompt to insist on grounding in retrieved context

- C. Disable evaluation
- D. Switch off streaming

Correct Answer: B. Improve retrieval quality (chunking, embeddings, hybrid search) and tighten the system prompt to insist on grounding in retrieved context

Explanation: Hallucinations in RAG are usually caused by weak retrieval (no relevant context) or weak instructions. Improving retrieval (chunking, embedding choice, hybrid + semantic search) and tightening the system prompt to demand citations and refusal when context is missing is the most effective combination.

Question 59

Which Foundry capability supports model evaluation against a labeled test set with both built-in (groundedness, relevance, coherence, fluency, F1) and custom evaluators?

✓ A. Foundry Evaluations

- B. Azure SQL Backups
- C. Azure ExpressRoute
- D. Azure Front Door

Correct Answer: A. Foundry Evaluations

Explanation: Foundry Evaluations let you run built-in (groundedness, relevance, coherence, fluency, F1, similarity, plus risk and safety) and custom evaluators against a labeled test set. Results are stored in the project for comparison across model and prompt versions.

Question 60

Which OpenAI Whisper-derived service in Azure transcribes audio at scale via a batch transcription REST API?

✓ **A. Azure AI Speech batch transcription**

- B. Azure Notification Hubs
- C. Azure Front Door
- D. Azure Service Bus

Correct Answer: A. Azure AI Speech batch transcription

Explanation: Azure AI Speech provides batch transcription that ingests audio files from blob storage and returns transcripts asynchronously. Azure OpenAI also exposes Whisper for short audio. For large-scale batch transcription, Azure AI Speech is the typical choice.

Question 61

Which Foundry feature lets developers iterate on a flow locally, then publish a new version to a Foundry project from CI/CD?

✓ **A. Prompt flow CLI / Foundry CLI commands invoked from GitHub Actions**

- B. Manual portal click only
- C. Azure DNS
- D. Azure Bastion

Correct Answer: A. Prompt flow CLI / Foundry CLI commands invoked from GitHub Actions

Explanation: Prompt flow / Foundry CLI commands (pf flow build, pf run create, az ml ... for endpoints) can be invoked from GitHub Actions to test, evaluate, and deploy flows. This enables proper CI/CD for prompt flows.

Question 62

Which Foundry construct contains chat history, run state, and tool execution context for an ongoing conversation with an agent?

✓ **A. Thread**

- B. Hub
- C. Registry
- D. Subscription

Correct Answer: A. Thread

Explanation: A thread holds the messages and state for an ongoing conversation with an agent. A run executes the agent against a thread, and the service manages tool calls and message persistence within the thread.

Question 63

Which approach correctly handles retries on 429 rate-limit errors from Azure OpenAI?

A. Immediate retry in a tight loop

✓ **B. Exponential backoff with jitter, respecting the Retry-After header**

- C. Fail fast and never retry
- D. Increase temperature on retry

Correct Answer: B. Exponential backoff with jitter, respecting the Retry-After header

Explanation: Azure OpenAI 429 responses include a Retry-After header. Clients should respect it and use exponential backoff with jitter to avoid thundering herds. The Azure SDKs implement this by default; if you write a custom HTTP client, you must implement it yourself.

Question 64

Which design pattern uses a smaller cheaper model to draft and a larger model to verify, balancing cost and quality?

✓ **A. Speculative decoding / draft-and-verify cascade**

- B. Random model selection
- C. Always use the largest model
- D. Always use the smallest model

Correct Answer: A. Speculative decoding / draft-and-verify cascade

Explanation: A cascade pattern uses a small model to draft an answer and a larger model to verify or rewrite. Speculative decoding is a related inference-time technique. Both reduce average cost while maintaining quality on hard cases.

Question 65

Which capability lets you customize an Azure OpenAI text-to-speech voice with sample recordings to create a brand voice?

✓ **A. Custom Neural Voice (Azure AI Speech)**

- B. Custom Vision
- C. Personalizer
- D. Custom Translator

Correct Answer: A. Custom Neural Voice (Azure AI Speech)

Explanation: Custom Neural Voice in Azure AI Speech lets you create a unique synthetic voice using approved voice talent recordings. Access requires application and approval due to responsible-AI considerations including consent and disclosure requirements.

Question 66

Which Azure AI Search vector configuration controls the algorithm and parameters used to build the approximate-nearest-neighbor index?

✓ **A. Vector profile referencing a vector search algorithm configuration (e.g., HNSW with m and efConstruction)**

- B. Storage account redundancy setting
- C. Network rule
- D. Replica count only

Correct Answer: A. Vector profile referencing a vector search algorithm configuration (e.g., HNSW with m and efConstruction)

Explanation: In Azure AI Search, vector profiles reference a vector search algorithm configuration (typically HNSW with m, efConstruction, efSearch parameters, plus an optional vectorizer). Tuning these affects index build cost, query latency, and recall.

Question 67

Which approach is required for an enterprise-grade Foundry deployment that must keep all traffic on the Microsoft backbone and prevent data exfiltration?

✓ **A. Use private endpoints for the hub, project, OpenAI, AI Search, storage, key vault inside a VNet**

- B. Open all firewall rules
- C. Use only the public Internet
- D. Run on a personal laptop

Correct Answer: A. Use private endpoints for the hub, project, OpenAI, AI Search, storage, key vault inside a VNet

Explanation: Configuring private endpoints for every dependent service inside a VNet ensures traffic stays on the Microsoft backbone with private IPs. Combined with disabling public network access, this is the recommended enterprise pattern for sensitive workloads.

Question 68

Which Azure OpenAI feature lets you store a system prompt and tool definitions as part of an Assistant/Agent so each new conversation starts pre-configured?

✓ **A. Agent (Assistant) definition with instructions, tools, and model**

- B. DNS records
- C. Public IP address
- D. Storage container ACL

Correct Answer: A. Agent (Assistant) definition with instructions, tools, and model

Explanation: An agent (Assistant) definition stores instructions (system prompt), allowed tools, and the model and parameters. New threads created against the agent inherit this configuration, ensuring consistent behavior without re-specifying it per request.

Question 69

Which approach is best to implement memory across sessions for a personal AI assistant in Foundry?

✓ **A. Persist user-specific facts to a datastore and inject them into the system prompt at the start of each new thread**

- B. Rely on the model's training data alone
- C. Store memory in plain DNS TXT records
- D. Hardcode memories in the system prompt

Correct Answer: A. Persist user-specific facts to a datastore and inject them into the system prompt at the start of each new thread

Explanation: Cross-session memory is implemented by persisting user-specific facts to a datastore (e.g., Cosmos DB, Azure SQL, Azure AI Search) keyed by user ID, then injecting relevant memories into the system prompt of new threads. This is sometimes called a memory store pattern.

Question 70

Which Azure capability lets you replay a captured request/response in Foundry tracing to debug a specific user-reported issue?

✓ **A. Tracing UI with end-to-end span details and replay/inspection**

- B. Azure DNS
- C. Azure Backup vault
- D. Azure CDN edge cache

Correct Answer: A. Tracing UI with end-to-end span details and replay/inspection

Explanation: Foundry's tracing UI surfaces full traces of a request including spans for retrieval, prompt, LLM call, and tools, with inputs/outputs at each step. Combined with App Insights logs, this lets engineers reconstruct, inspect, and replay specific user-reported issues.

Question 71

Which Azure OpenAI deployment property controls how many tokens per minute (TPM) clients can send to a deployment before being throttled?

✓ **A. TPM quota assigned to the deployment**

- B. Storage redundancy
- C. Region
- D. Model name

Correct Answer: A. TPM quota assigned to the deployment

Explanation: Each Azure OpenAI deployment is allocated a TPM (tokens per minute) quota out of the regional capacity pool. Clients exceeding TPM (or RPM, requests per minute) receive 429 throttling responses. Quota can be requested via support or self-service in the portal.

Question 72

Which approach lets you efficiently send the same long context (e.g., a static knowledge document) across many requests without paying full token cost each time?

✓ **A. Prompt caching (when supported by the model and SDK)**

- B. Increase temperature
- C. Use embeddings
- D. Use Whisper

Correct Answer: A. Prompt caching (when supported by the model and SDK)

Explanation: Prompt caching (supported on selected Azure OpenAI models) detects identical prompt prefixes across requests and reuses cached computation, lowering both latency and per-request cost for the cached portion. This is ideal for repeated long system prompts.

Question 73

Which Azure AI Document Intelligence model best identifies the type of an unknown document (invoice, receipt, contract, etc.) before downstream extraction?

✓ **A. Custom classification model**

- B. Read API
- C. Translator
- D. Speech to text

Correct Answer: A. Custom classification model

Explanation: Custom classification models in Azure AI Document Intelligence are trained on labeled samples to assign incoming documents to one of several classes. The classification result then drives which extraction model is used downstream.

Question 74

Which Foundry concept represents a single environment (typically per team or per workload) in which AI applications are developed and deployed?

✓ **A. Project**

- B. Subscription
- C. Resource group
- D. Region

Correct Answer: A. Project

Explanation: A Foundry project sits under a hub and is the unit where teams develop and deploy AI apps and agents. Projects own deployments, indexes, connections, datasets, and evaluation results, while inheriting shared infra from the hub.

Question 75

Which Azure AI service is the recommended starting point for performing real-time speech translation across many languages?

✓ A. Speech translation in Azure AI Speech

- B. Custom Vision
- C. Personalizer
- D. Anomaly Detector

Correct Answer: A. Speech translation in Azure AI Speech

Explanation: Azure AI Speech provides speech translation that combines speech-to-text, machine translation, and (optionally) text-to-speech for real-time multilingual conversations. It is the recommended Azure-native approach for spoken translation scenarios.

Question 76

Which Azure AI Vision feature analyzes image content and returns a list of detected objects with bounding boxes?

✓ A. Object detection

- B. Speaker recognition
- C. QnA Maker
- D. Personalizer

Correct Answer: A. Object detection

Explanation: Object detection in Azure AI Vision returns a list of detected objects with class labels and bounding box coordinates. For domain-specific objects, train a custom Image Analysis model in Vision Studio with labeled images.

Question 77

Which approach to building a domain-specific image classifier requires the smallest amount of labeled training data?

A. Train a model from scratch with millions of images

✓ B. Train a custom model in Azure AI Vision Image Analysis using transfer learning on the prebuilt foundation model

- C. Use a pure Python script with no pretrained model
- D. Use only data augmentation

Correct Answer: B. Train a custom model in Azure AI Vision Image Analysis using transfer learning on the prebuilt foundation model

Explanation: Custom Image Analysis in Azure AI Vision uses transfer learning from a pretrained foundation model, so a small labeled dataset (often dozens to hundreds of images per class) is sufficient. Training from scratch requires orders of magnitude more data.

Question 78

Which Foundry capability lets you ground an agent on a curated set of websites, returning answers with web citations?

✓ A. Bing/Web Grounding tool for agents

- B. Azure DNS
- C. Azure Cosmos DB
- D. Azure Storage

Correct Answer: A. Bing/Web Grounding tool for agents

Explanation: The Web Grounding (Bing) tool for Foundry agents allows the agent to perform web search and use the results as grounding, returning citations to the user. It is intended for use cases where current public web content is needed.

Question 79

Which Azure OpenAI capability is the most appropriate choice for generating images from text prompts in 2026?

✓ **A. DALL-E 3 (or successor)**

- B. Whisper
- C. text-embedding-3-small
- D. gpt-4o-mini

Correct Answer: A. DALL-E 3 (or successor)

Explanation: DALL-E 3 (and its successors) is the Azure OpenAI image-generation model. It generates images from text prompts and is appropriate for creative and design scenarios, subject to content policies and content filters.

Question 80

Which Azure feature lets you store secrets (API keys, connection strings) used by a Foundry-powered app in a secure, RBAC-protected vault?

✓ **A. Azure Key Vault**

- B. Azure Storage table
- C. Azure DNS TXT records
- D. Azure Front Door

Correct Answer: A. Azure Key Vault

Explanation: Azure Key Vault is the standard Azure secrets store. Apps retrieve secrets at runtime via managed identity with RBAC, eliminating embedded secrets in code or config files. Foundry hubs are linked to a Key Vault for credentials of connections.

Question 81

Which Foundry capability lets you compare two model versions in production by routing 90% of traffic to the current model and 10% to the new one?

✓ **A. Traffic split (canary) on a managed online endpoint**

- B. Single deployment only
- C. Manual DNS swap
- D. VPN failover

Correct Answer: A. Traffic split (canary) on a managed online endpoint

Explanation: Managed online endpoints support multiple deployments behind a single endpoint with weighted traffic routing. A 90/10 split is a canary that exposes a small percentage of users to the new model so you can compare metrics before promoting.

Question 82

Which evaluator should you choose to detect harmful content (hate, sexual, violence, self-harm) in generated responses?

✓ **A. Risk and safety evaluators (Azure AI Content Safety harm models)**

- B. Coherence evaluator

- C. Latency
- D. Throughput

Correct Answer: A. Risk and safety evaluators (Azure AI Content Safety harm models)

Explanation: Foundry's risk and safety evaluators wrap Azure AI Content Safety harm classification models to score outputs for hate, sexual, violence, and self-harm severity, plus jailbreak attack and protected material detection.

Question 83

Which Foundry tool lets an agent execute long-running workflows by calling Azure Logic Apps as an OpenAPI-compatible tool?

✓ A. OpenAPI tool (or Logic Apps connector tool)

- B. Azure Backup tool
- C. Azure Bastion tool
- D. Azure Monitor tool

Correct Answer: A. OpenAPI tool (or Logic Apps connector tool)

Explanation: Foundry agents can call OpenAPI-described external endpoints, including Azure Logic Apps, as tools. The agent invokes the API by name and the tool's OpenAPI spec lets the model construct correctly typed arguments.

Question 84

Which Foundry feature ensures that an agent only consumes models, indexes, and connections defined in its current project (preventing access to other projects)?

✓ A. Project-scoped resources and RBAC isolation

- B. Public endpoints
- C. Anonymous access
- D. Disable RBAC

Correct Answer: A. Project-scoped resources and RBAC isolation

Explanation: Foundry scopes model deployments, indexes, datasets, and connections to a project, with RBAC enforced at the project level. An agent can only consume resources within its project, isolating teams and workloads.

Question 85

Which method should you use to evaluate a flow's quality on every pull request in CI?

✓ A. Run pf flow test or batch evaluations against an evaluation dataset in GitHub Actions and fail PRs that regress thresholds

- B. Skip evaluation in CI
- C. Have a developer eyeball it once
- D. Wait until production complaints

Correct Answer: A. Run pf flow test or batch evaluations against an evaluation dataset in GitHub Actions and fail PRs that regress thresholds

Explanation: Continuous evaluation in CI (running prompt flow tests and bulk evaluations against a labeled dataset on every PR) catches regressions before merge. Configure thresholds for groundedness, relevance, safety, etc., and fail PRs that breach them.

Question 86

Which Foundry capability lets you deploy a prompt flow as a managed online endpoint that other apps can call as a REST API?

✓ A. Deploy flow to a managed online endpoint

- B. Disable network access
- C. Save flow as PDF
- D. Email flow YAML

Correct Answer: A. Deploy flow to a managed online endpoint

Explanation: After developing and testing a prompt flow, you can deploy it as a managed online endpoint. This packages the flow, dependencies, and configuration into a containerized inference service exposed as a REST API that other applications consume.

Question 87

Which approach should you use to ensure that an Azure OpenAI deployment used by your app is in the same region as your data, for both latency and data-residency reasons?

✓ **A. Choose the deployment region to match the storage region and configure private endpoints in that region**

- B. Always use a different region
- C. Use a free tier in any region
- D. Disable region selection

Correct Answer: A. Choose the deployment region to match the storage region and configure private endpoints in that region

Explanation: Co-locating Azure OpenAI deployments and dependent resources (storage, AI Search, app compute) in the same region minimizes latency, reduces egress cost, and simplifies data-residency compliance. Use private endpoints for backbone-only traffic.

Question 88

Which Azure AI capability identifies known faces (e.g., employees) in images using a Limited Access enrolled face list?

✓ **A. Face Recognition (Azure AI Face) with Limited Access controls**

- B. Translator
- C. QnA Maker
- D. Speaker Identification

Correct Answer: A. Face Recognition (Azure AI Face) with Limited Access controls

Explanation: Azure AI Face supports face recognition using enrolled face lists. Identify, verify, and group operations are gated behind Microsoft's Limited Access program due to responsible-AI considerations. Speaker Identification recognizes voices, not faces.

Question 89

Which Microsoft Foundry / Azure AI service is intended to extract insights from unstructured documents combining text, images, and structure (formerly known as Azure AI Studio Content Understanding)?

✓ **A. Content Understanding (Azure AI Content Understanding)**

- B. Custom Vision
- C. Anomaly Detector
- D. Personalizer

Correct Answer: A. Content Understanding (Azure AI Content Understanding)

Explanation: Azure AI Content Understanding extracts insights and structured data from documents that combine text, images, audio, or video using configurable analyzers. It is increasingly the preferred way to extract structured information from rich, multi-modal content within Foundry workflows.

Question 90

Which Foundry feature collects user thumbs-up/thumbs-down feedback on generated responses for offline analysis?

✓ **A. User feedback capture in the SDK / endpoint, surfaced in Foundry monitoring**

- B. Azure DNS
- C. Azure Bastion
- D. Azure Backup

Correct Answer: A. User feedback capture in the SDK / endpoint, surfaced in Foundry monitoring

Explanation: Foundry SDKs and managed online endpoints support capturing user feedback signals (thumbs-up/down, ratings, comments) and storing them with request IDs. The signals can be analyzed in Foundry monitoring or exported to Application Insights / Log Analytics.

Question 91

Which Foundry feature lets you run the same evaluation against multiple deployed models to choose the best one for your task?

✓ **A. Model evaluation comparison across deployments using a shared evaluation dataset**

- B. Random selection
- C. Highest-cost model wins
- D. Whichever name sounds best

Correct Answer: A. Model evaluation comparison across deployments using a shared evaluation dataset

Explanation: Foundry evaluations can target multiple deployments with the same dataset and evaluators, producing comparable metrics for each. This is the principled way to select among models (e.g., gpt-4.1 mini vs gpt-4o vs an open-source model) for your workload.

Question 92

Which Azure AI Speech capability provides low-latency, full-duplex voice conversations suitable for voice agents?

✓ **A. Voice Live (real-time conversational voice)**

- B. Batch transcription
- C. Custom Translator
- D. Personalizer

Correct Answer: A. Voice Live (real-time conversational voice)

Explanation: Voice Live (real-time conversational voice) provides low-latency, full-duplex voice interaction by combining streaming speech-to-text, an LLM, and streaming text-to-speech. It targets voice agent and live conversational scenarios.

Question 93

Which Azure AI Search capability lets you ingest and serve queries against vectors generated outside the index (e.g., by your own embedding pipeline)?

✓ **A. Pre-computed vector fields populated by the developer at index time**

- B. Mandatory built-in embedding only
- C. Vector indexing is unsupported
- D. Only keyword search

Correct Answer: A. Pre-computed vector fields populated by the developer at index time

Explanation: Azure AI Search supports pre-computed vectors uploaded as part of documents. You can also use integrated vectorization to compute vectors at index time. Both options coexist, allowing flexibility based on your embedding pipeline.

Question 94

Which Foundry concept is the JSON-defined evaluation result of running a flow against a dataset, capturing metrics per row and aggregate?

✓ **A. Run (evaluation run)**

- B. Subscription
- C. Storage container
- D. Region

Correct Answer: A. Run (evaluation run)

Explanation: An evaluation run in Foundry executes a flow against an evaluation dataset and applies one or more evaluators. The run record includes per-row inputs, outputs, evaluator scores, and aggregate metrics, all viewable in the studio and queryable via the SDK.

Question 95

Which approach should you use to give a Foundry agent access to internal company REST APIs without exposing them publicly?

✓ **A. Expose APIs through a private endpoint or Azure API Management with managed identity authentication, then register them as agent tools**

- B. Make all internal APIs public
- C. Email the model the API specs
- D. Disable authentication on internal APIs

Correct Answer: A. Expose APIs through a private endpoint or Azure API Management with managed identity authentication, then register them as agent tools

Explanation: Internal APIs should remain private (private endpoints, VNet access, or fronted by Azure API Management). Register them as agent tools via OpenAPI specs, with the agent's compute using managed identity to authenticate. APIM can also enforce throttling and policy.

Question 96

Which Foundry capability lets you specify per-deployment content filters that differ from the default (e.g., higher tolerance for medical terminology in a healthcare app)?

✓ **A. Custom content filter applied to a specific Azure OpenAI deployment**

- B. Disable all filters globally
- C. Use Azure DNS
- D. Use Azure Bastion

Correct Answer: A. Custom content filter applied to a specific Azure OpenAI deployment

Explanation: Custom content filters in Azure OpenAI can be created and assigned to specific deployments, allowing per-deployment severity thresholds and optional add-ons (Prompt Shields, protected material). This enables workload-specific tuning while keeping safety controls in place.

Question 97

Which approach should you use to detect when a deployed prompt flow's groundedness scores degrade over time?

✓ **A. Continuous evaluation with groundedness evaluator and Azure Monitor alert on threshold**

- B. Disable monitoring
- C. Wait for users to complain
- D. Re-train weekly without measurement

Correct Answer: A. Continuous evaluation with groundedness evaluator and Azure Monitor alert on threshold

Explanation: Continuous evaluation samples production traffic and applies the groundedness evaluator on a schedule. Configuring an Azure Monitor alert on the resulting metric when it falls below a threshold creates an automated quality regression detector.

Question 98

Which Azure OpenAI capability provides a structured way to ground responses in chat history while using minimal context window (e.g., summarizing prior turns)?

- ✓ **A. Conversation summarization (manual or via tool that summarizes older turns)**
- B. Increase max_tokens to 1 million
- C. Use embeddings only
- D. Disable system prompt

Correct Answer: A. Conversation summarization (manual or via tool that summarizes older turns)

Explanation: Long conversations exceed practical context windows. Implementing conversation summarization (replacing older turns with a summary, generated by the LLM itself) preserves continuity while keeping the prompt within budget. Memory stores can complement this for cross-session memory.

Question 99

Which Foundry deployment artifact must be committed to source control to make a flow reproducible?

- ✓ **A. flow.dag.yaml plus requirements.txt and node implementation files (Python/Jinja)**
- B. Only screenshots
- C. Only the binary state file
- D. An exported PDF

Correct Answer: A. flow.dag.yaml plus requirements.txt and node implementation files (Python/Jinja)

Explanation: A reproducible Foundry flow requires its definition (flow.dag.yaml), Python and Jinja node implementations, and pinned dependencies (requirements.txt) committed to Git. CI/CD then builds, evaluates, and deploys from a known commit.

Question 100

Which combined practice best characterizes the AI-103 audience profile's responsibilities?

- A. Designing UI/UX for marketing pages
- ✓ **B. Building AI-infused applications and agents on Microsoft Foundry, integrating Azure OpenAI, Azure AI Search, vision, speech, language, and content safety, while applying responsible AI practices**
- C. Writing GPU drivers
- D. Configuring on-premises Active Directory

Correct Answer: B. Building AI-infused applications and agents on Microsoft Foundry, integrating Azure OpenAI, Azure AI Search, vision, speech, language, and content safety, while applying responsible AI practices

Explanation: Per the AI-103 audience profile, candidates build AI-infused applications and agents on Microsoft Foundry, leveraging Azure OpenAI, Azure AI Search, vision, speech, language, document intelligence, and content

safety, with Python or C# and Azure SDKs, while applying responsible AI practices.

Section 2 — Additional AI-103 Questions (Community Exam-Prep Sources)

Sourced from public AI-103 exam-prep sites (CertyIQ, Pass4Success, ITExams). Answers marked [source-confirmed] were provided by the source's answer key; answers marked [expert-verified] were determined against Microsoft Learn documentation. HOTSPOT/DRAG-DROP/case-study items were excluded because their answer keys are image-based and do not fit a 4-option format.

Question 1

An AI-103 case-study agent (Agent1) ingests product sheets that may contain images with embedded text. The agent must be protected from malicious instructions potentially hidden within those images. Which Azure AI Content Safety capability should you use?

- A. Self-harm content filtering
- ✓ B. Prompt Shields
- C. Personally identifiable information (PII) Detection
- D. Violence content filtering

Correct Answer: B. Prompt Shields

Explanation: Prompt Shields detect direct jailbreak attempts and indirect (cross-prompt) injection attacks, including malicious instructions hidden in retrieved content and multimodal inputs such as images. [source-confirmed: CertyIQ AI-103 Q2]

Source: CertyIQ AI-103 Q2

Question 2

A Microsoft Foundry project (Project1) contains multiple agents that each access the same Azure AI Search resource. You need to centrally manage the Azure AI Search credentials so the solution applies across all agents. What should you recommend?

- A. Enable role-based access control (RBAC) for the Azure AI Search resource
- B. Disable key-based access control on the Azure AI Search resource
- ✓ C. Add a connection to the Azure AI Search resource
- D. Create a managed private endpoint that connects to the Azure AI Search resource

Correct Answer: C. Add a connection to the Azure AI Search resource

Explanation: In Microsoft Foundry, a project-level connection centrally stores and manages credentials for external resources (Azure AI Search, Azure OpenAI, Storage, etc.) so every agent in the project can reuse the same connection. [source-confirmed: CertyIQ AI-103 Q3]

Source: CertyIQ AI-103 Q3

Question 3

You need to orchestrate three Foundry agents using a deterministic, step-based process that uses conditional branching and shared state, optionally triggering a ticket action based on the triage result, with minimal development effort. What should you include in the solution?

- ✓ A. A workflow
- B. Threads and runs without a workflow
- C. A multi-agent group chat session
- D. Separate agent runs coordinated in the application code

Correct Answer: A. A workflow

Explanation: A workflow lets you declaratively define ordered agent steps, pass shared state between steps, apply conditional branching, and invoke actions/tools based on outcomes — orchestrating multiple agents with minimal custom code. [source-confirmed: CertylQ/ITExams AI-103 Q9]

Source: CertylQ / ITExams AI-103 Q9

Question 4

An agent uses Azure Speech in Foundry Tools with a fine-tuned, published custom speech-to-text model for the en-us locale. The agent calls the Speech to text REST API and receives an error indicating the project ID is invalid. To what should you set the project property?

- A. The project URL
- ✓ B. The custom speech project ID
- C. The project ID
- D. The custom speech endpoint URL

Correct Answer: B. The custom speech project ID

Explanation: When using a custom Speech-to-Text model, the project property must reference the Custom Speech project ID that contains the trained model — which is distinct from the Foundry project ID. [source-confirmed: CertylQ AI-103 Q10]

Source: CertylQ AI-103 Q10

Question 5

An agent uses a custom speech-to-text model deployed to a custom endpoint for real-time recognition. You are approaching the model's expiration date. What is the expected behavior when the custom model expires?

- A. Speech recognition requests return a 4xx error until a new custom model is deployed
- B. Speech recognition requests continue using the expired custom model until it is removed manually
- ✓ C. Speech recognition requests fall back to the most recent base model for the same locale
- D. The custom model is deleted automatically when it expires

Correct Answer: C. Speech recognition requests fall back to the most recent base model for the same locale

Explanation: On expiration, the endpoint keeps serving requests but automatically falls back to the latest Microsoft base model for the same locale, preserving service continuity (though customization quality may differ).

[source-confirmed: CertylQ AI-103 Q12]

Source: CertylQ AI-103 Q12

Question 6

Developers calling an Azure OpenAI deployment with the Azure OpenAI v1 API and DefaultAzureCredential receive HTTP 403 errors when sending inference requests, even after running az login. Following least privilege, which RBAC role should you assign so they can perform model inference?

- A. Cognitive Services User
- ✓ B. Cognitive Services OpenAI User
- C. Contributor
- D. Cognitive Services Data Reader

Correct Answer: B. Cognitive Services OpenAI User

Explanation: The Cognitive Services OpenAI User role grants the minimum data-plane permissions required to invoke model deployments (chat completions, embeddings, inference) and follows the principle of least privilege.

[source-confirmed: CertylQ AI-103 Q13]

Source: CertylQ AI-103 Q13

Question 7

An agent has a Model Context Protocol (MCP) tool that queries a knowledge base in Azure AI Search, but some runs answer from the base model without grounded citations. Which tool_choice value deterministically forces the agent to invoke the MCP tool on every run?

✓ **A. tool_choice={"required"}**

B. tool_choice={"auto"}

C. tool_choice={"type":"knowledge_base"}

D. tool_choice={"type":"mcp"}

Correct Answer: A. tool_choice={"required"}

Explanation: Setting tool_choice to "required" forces the model to invoke at least one tool on every run, removing its option to respond with plain text and ensuring the tool (e.g., the MCP knowledge-base tool) is always called.

[source-confirmed: CertylQ AI-103 Q14]

Source: CertylQ AI-103 Q14

Question 8

A Foundry agent calls an external weather service via an OpenAPI 3.0 specification that requires a key passed in an HTTP header. The key value is stored as a connection in Project1. What should you configure in the OpenAPI specification so the key is included automatically whenever the tool is invoked?

A. A header parameter defined for each operation

B. An Azure Key Vault connection

✓ **C. An API key security scheme**

D. A Bearer token security scheme

Correct Answer: C. An API key security scheme

Explanation: An API key security scheme in the OpenAPI spec tells the Foundry agent to automatically inject the connection's stored key into the configured HTTP header on each call. [source-confirmed: CertylQ AI-103 Q16]

Source: CertylQ AI-103 Q16

Question 9

A Foundry project serves a high-volume chat app where most requests are simple FAQs but some require advanced reasoning. You must reduce cost and latency for common queries without degrading the quality of responses to complex questions. What should you do?

A. Route all the requests to a smaller model

✓ **B. Use a model cascade that routes the requests to different models**

C. Increase the value of the max_tokens parameter for all the requests

D. Route all the requests to the most capable model

Correct Answer: B. Use a model cascade that routes the requests to different models

Explanation: A model cascade routes simple, common queries to a small low-cost/low-latency model and escalates only complex queries to a more capable model, optimizing cost and latency while preserving quality where it matters.

[source-confirmed: CertylQ AI-103 Q17]

Source: CertylQ AI-103 Q17

Question 10

A high-traffic agent's operational costs increase significantly after an update, while user traffic volume is unchanged. You need to determine whether the additional cost is driven by model input size, model output size, or expanded tool usage. Which observability capability should you use?

- A. Latency
- B. Evaluation metrics
- C. Run success rate

✓ D. Token usage

Correct Answer: D. Token usage

Explanation: Token usage breaks down input (prompt) tokens, output (completion) tokens, and tool-call token consumption per run — the direct drivers of token-based model cost. [source-confirmed: CertyIQ/Pass4Success AI-103 Q19]

Source: CertyIQ / Pass4Success AI-103 Q19

Question 11

You are deploying a support agent that lets users upload photos. You need to automatically classify uploaded images for harmful content and block content based on severity levels. What should you do?

✓ A. Implement image moderation

- B. Enable Prompt Shields
- C. Apply keyword scanning to OCR output by using Azure Vision in Foundry Tools
- D. Use blocklists

Correct Answer: A. Implement image moderation

Explanation: Azure AI Content Safety image moderation classifies images across harm categories (hate, sexual, violence, self-harm) with severity levels and can block content that exceeds a configured threshold. Prompt Shields target prompt-injection, and blocklists/keyword scanning target text. [expert-verified against Microsoft Learn; Pass4Success AI-103 Q3]

Source: Pass4Success AI-103 Q3

Question 12

A chat app in a Foundry project uses an Azure AI Search vectorized index. Retrieval must pull information from multiple chunks for complex questions, let multi-turn conversation influence retrieval planning, and run retrievals in parallel to reduce latency. Which retrieval approach should you use?

- A. Classic Retrieval Augmented Generation (RAG)
- B. Chain of thought

✓ C. Agentic Retrieval Augmented Generation (RAG)

- D. Iterative retrieval

Correct Answer: C. Agentic Retrieval Augmented Generation (RAG)

Explanation: Agentic RAG decomposes a complex query into multiple subqueries, uses conversation history to plan retrieval, and runs those retrievals in parallel before synthesizing an answer — matching all three requirements. [expert-verified against Microsoft Learn; Pass4Success AI-103 Q4]

Source: Pass4Success AI-103 Q4

Question 13

You build a real-time voice agent that must receive continuous audio input, convert it to text for reasoning, support turn-taking (begin generating speech before the user finishes speaking), operate at low latency, and return spoken responses. What should you do?

- A. Use an embeddings model to encode the audio, then decode the audio into text and speech
- B. Use batch transcription to convert the audio input and return text responses
- C. Use speech translation to convert the audio into another language and return the translated text
- ✓ D. Use real-time speech to text for incoming audio and text to speech for agent responses

Correct Answer: D. Use real-time speech to text for incoming audio and text to speech for agent responses

Explanation: Real-time (streaming) speech-to-text for incoming audio combined with text-to-speech for responses provides the low-latency, bidirectional, turn-taking voice experience required. Batch transcription and speech translation do not meet the real-time/turn-taking requirements. [expert-verified against Microsoft Learn; Pass4Success AI-103 Q5]

Source: Pass4Success AI-103 Q5
